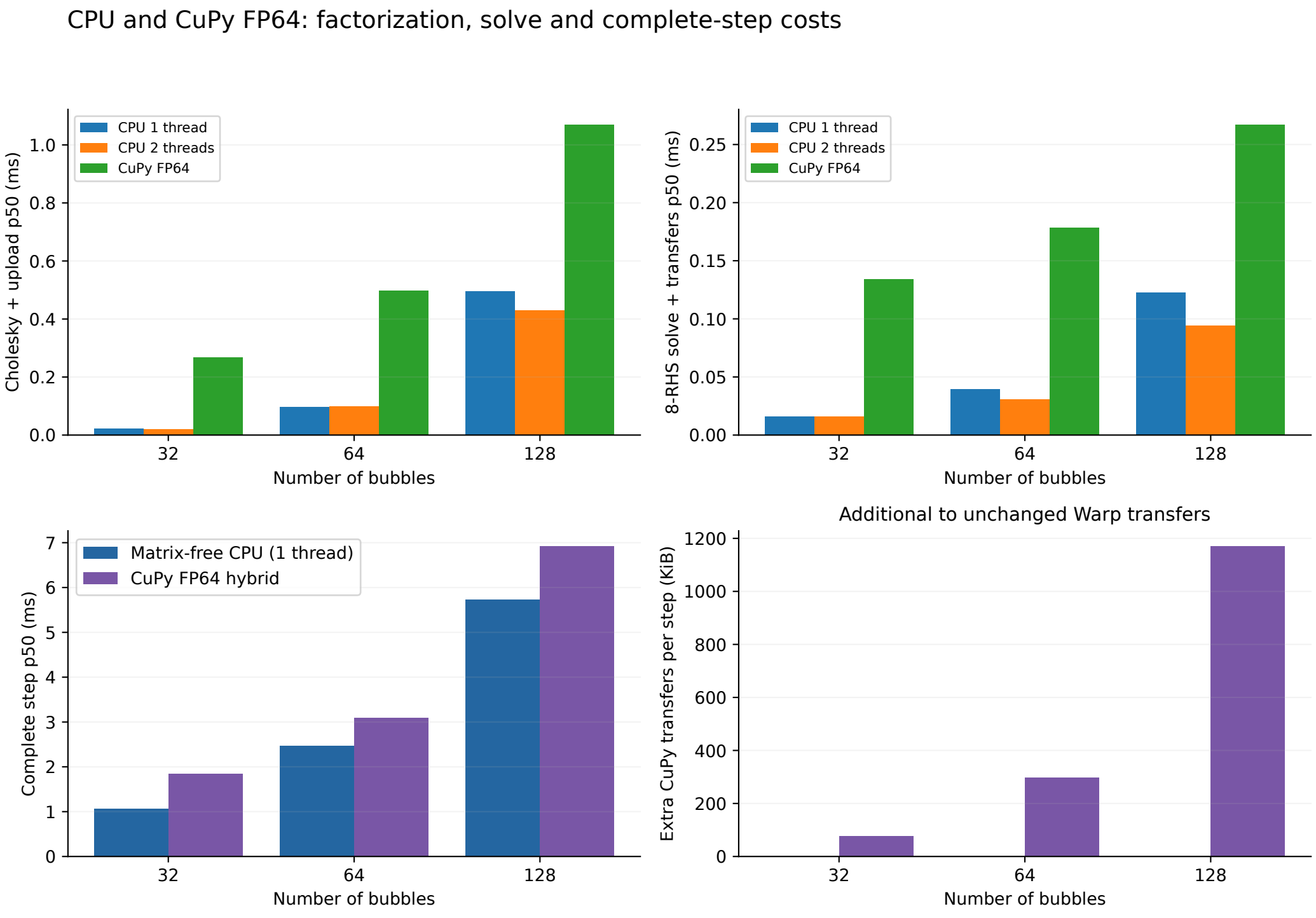
CPU and CuPy FP64: factorization, solve and complete-step costs
CPU 1 thread
CPU 2 threads
CuPy FP64
Cholesky + upload p50 (ms)
0.0
0.2
0.4
0.6
0.8
1.0
32
64
128
Number of bubbles
CPU 1 thread
CPU 2 threads
CuPy FP64
8-RHS solve + transfers p50 (ms)
0.00
0.05
0.10
0.15
0.20
0.25
32
64
128
Number of bubbles
Matrix-free CPU (1 thread)
CuPy FP64 hybrid
Complete step p50 (ms)
0
1
2
3
4
5
6
7
32
64
128
Number of bubbles
Additional to unchanged Warp transfers
Extra CuPy transfers per step (KiB)
0
200
400
600
800
1000
1200
32
64
128
Number of bubbles
EXPLORATORY | Uncalibrated physics | Homogeneous-water proxy | training_ready=false